

尺寸：220×325(mm) 材质：70g双面胶版纸

Fentwell 0.1 mg/2 mL Solution for Injection

Fentanyl Citrate Injection

0.1571mg Fentanyl Citrate equivalent to 0.1mg Fentanyl

PRODUCT NAME

Generic Name: Fentanyl Citrate Injection

Brand Name: Fentwell

COMPOSITION

Active substance: Fentanyl Citrate

The chemical name is

N-(1-phenethyl-4-piperidyl)

propionanilide citrate (1:1).

The molecular weight is 528.60.

Its molecular formula is

C₂₂H₂₆N₂O • C₈H₈O₄,

and it has the following chemical structure.

Fentanyl citrate 78.5 micrograms equivalent to 50 micrograms per ml fentanyl.

Each 2ml ampoule contains 100 micrograms of fentanyl as fentanyl citrate.

Strength of Active substance: 0.1571mg Fentanyl Citrate equivalent to 0.1mg Fentanyl

Excipients: Sodium Hydroxide, Water for Injection

PRODUCT DESCRIPTION

This product is a sterile, non-pyrogenic, preservative free, clear and colorless aqueous solution for intravenous use.

The product can be further diluted with sodium chloride or glucose injection.

After dilution with the diluents, the solution appears to be clear and colorless.

CLINICAL PHARMACOLOGY

Pharmacodynamics

Pharmacotheapeutic group: Opioid analgesic, ATC code: N01AH01.

Mechanism of action

Fentanyl is a potent, opioid analgesic.

Pharmacodynamic effects

Fentanyl is an opioid analgesic interacting predominantly with the μ -opioid receptor. Fentanyl can be used as an analgesic supplement to general anesthesia or as the sole anesthetic. Fentanyl preserves cardiac stability and obtunds stress-related hormonal changes at higher doses. A dose of 100 mcg (2.0 mL) is approximately equivalent in analgesic activity to 10 mg of morphine. The onset of action is rapid. However, the maximum analgesic and respiratory depressant effect may not be noted for several minutes. The usual duration of action of the analgesic effect is approximately 30 minutes after a single IV dose of up to 100 mcg. Depth of analgesia is dose related and can be adjusted to the pain level of the surgical procedure.

Like other opioid analgesics, depending upon the dose and speed of administration, Fentanyl can cause muscle rigidity, as well as euphoria, miosis and bradycardia.

All actions of Fentanyl are reversed by a specific opioid antagonist.

Pharmacokinetics

Absorption

Fentanyl is a lipid-soluble drug and its pharmacokinetics can be described in terms of a three-compartment model. Following intravenous injection, there is a short distribution phase during which high concentrations of fentanyl are achieved quickly in well-perfused tissues such as the lungs, kidneys and brain.

Distribution

The drug is redistributed to other tissues; it accumulates more slowly in skeletal muscle and yet more slowly in fat, from which it is gradually released into the blood. Up to 80% of fentanyl is bound to plasma proteins.

Elimination

Fentanyl is primarily metabolised in the liver, probably by N-dealkylation, and it is excreted mainly in the urine with less than 10% representing the unchanged drug. The terminal half-life of fentanyl is 3.7 hours.

INDICATIONS

Fentanyl is indicated:

- for use as an opioid analgesic supplement in general or regional anesthesia.
- for administration with a neuroleptic as an anesthetic premedication, for the induction of anesthesia, and as an adjunct in the maintenance of general and regional anesthesia.
- for use as an anesthetic agent with oxygen in selected high-risk patients undergoing major surgery.

DOSAGE AND ADMINISTRATION

The dosage of fentanyl should be individualized according to age, body weight, physical status, underlying pathological condition, use of other drugs, and type of surgery and anesthesia.

To avoid bradycardia, it is recommended to administer a small intravenous dose of an anti-cholinergic just before anesthetic induction.

Use as an analgesic supplement to general anesthesia

Low dose: 2 mcg/kg.

Fentanyl in small doses is most useful for minor, surgery.

Moderate dose: 2-20 mcg/kg.

Where surgery becomes more complicated, a larger dose will be required. The duration of activity is dependent on dosage.

High dose: 20-50 mcg/kg.

During major surgical procedures, in which surgery is longer, and during which the stress response would be detrimental to the well-being of the patient, doses of 20-50 mcg/kg of fentanyl with nitrous oxide/oxygen have been shown to have an attenuating effect. When doses in this range have been used during surgery, post-operative ventilation and observation are essential in view of the possibility of extended post-operative respiratory depression.

Supplemental doses of 25-250 mcg (0.5-5 mL) should be tailored to the needs of the patient and to the anticipated time until completion of the operation.

Use as an anesthetic agent

When attenuation of the response to surgical stress is especially important, doses of 50-100 mcg/kg may be administered with oxygen and a muscle relaxant. This technique provides anesthesia without necessitating the use of additional anesthetic agents. In certain cases, doses up to 150 mcg/kg may be required to produce this anesthetic effect. Fentanyl has been used in this fashion for open heart surgery and certain other major surgical procedures in patients for whom protection of the myocardium from excess oxygen demand is particularly indicated.

Special Populations

Pediatrics

For induction and maintenance in children aged 2-12 years, a dose of 2-3 mcg/kg is recommended.

Elderly and debilitated patients

As with other opioids, the initial dose should be reduced in the elderly (> 65 years of age) and in debilitated patients. The effect of the initial dose should be taken into account in determining supplemental doses.

Obese patients

In obese patients there is a risk of overdosing if the dose is calculated based on body weight. Obese patients should be dosed based on estimated lean body mass rather than on body weight only.

Renal impairment

In patients with renal impairment reduced dosing of fentanyl should be considered and these patients should be observed carefully for signs of fentanyl toxicity (see Pharmacokinetic Properties).

ROUTE OF ADMINISTRATION

INTRAVENOUS (IV)

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients.

Known intolerance to fentanyl or other morphino-mimetics.

Respiratory depression.

Obstructive airways disease.

In patients after operative interventions in the biliary tract.

WARNINGS AND PRECAUTIONS

Fentanyl should be given only in an environment where the airway can be controlled and by personnel who can control the airway.

As with all potent opioids.

Respiratory depression is dose related and can be reversed by a specific narcotic antagonist such as naloxone, but additional doses of the latter may be necessary because the respiratory depression may last longer than the duration of action of the opioid antagonist. Profound analgesia is accompanied by marked respiratory depression, which can persist or recur in the postoperative period.

Therefore, patients should remain under appropriate surveillance. Resuscitation equipment and narcotic antagonists should be readily available. Hyperventilation during anaesthesia may alter the patient's response to CO₂, thus affecting respiration post-operatively. Induction of muscle rigidity, which may also involve the thoracic muscles, can occur, but can be avoided by the following measures: slow I.V. injection (ordinarily sufficient for lower doses), premedication with benzodiazepines and the use of muscle relaxants.

Non-epileptic (myo)clonic movement can occur.

Bradycardia, and possibly cardiac arrest, can occur if the patient has received an insufficient amount of anticholinergic, or when fentanyl is combined with non-vagolytic muscle relaxants. Bradycardia can be treated with atropine.

It is imperative to ensure that adequate spontaneous breathing has been established and maintained before discharge from the recovery area whenever large doses or infusions of Fentanyl 50 micrograms/mL Solution for Injection/Infusion have been administered.

Repeated use of fentanyl may result in the development of tolerance and dependence.

Opioids may induce hypotension, especially in hypovolemic patients. Appropriate measures to maintain a stable arterial pressure should be taken.

The use of rapid bolus injection of opioids should be avoided in patients with compromised intracerebral compliance; in such patient the transient decrease in the mean arterial pressure has occasionally been accompanied by a short-lasting reduction of the cerebral perfusion pressure. Patients on chronic opioid therapy or with a history of opioid abuse may require higher doses.

It is recommended to reduce dosage in the elderly and in debilitated patients.

Opioids should be titrated with caution in patients with any of the following conditions: uncontrolled hypothyroidism, pulmonary disease, decreased respiratory reserve, alcoholism or impaired renal or hepatic function. Such patients also require prolonged postoperative monitoring.

If fentanyl is administered with a neuroleptic [such as droperidol], the user should be familiar with the special properties of each drug, particularly the difference in duration of action. When such a combination is used, there is higher incidence of hypotension. Neuroleptics can induce extrapyramidal symptoms that can be controlled with anti-Parkinson agents.

As with other opioids, due to the anticholinergic effects, administration of fentanyl may lead to increases of bile duct pressure and, in isolated cases, spasms of the sphincter of Oddi might be observed.

In patients with myasthenia gravis, careful consideration should be applied in the use of certain anticholinergic agents and neuromuscular-blocking pharmaceutical agents prior to, and during, the administration of a general anesthetics regimen which includes administering intravenous fentanyl.

Administration of fentanyl during labour may result in neonatal respiratory depression.

Serotonin Syndrome

Caution is advised when fentanyl is coadministered with drugs that affect the serotonergic neurotransmitter systems.

The development of a potentially life-threatening serotonin syndrome may occur with the concomitant use of serotonergic drugs such as Selective Serotonin Re-uptake Inhibitors (SSRIs) and Serotonin Norepinephrine Re-uptake Inhibitors (SNRIs), and with drugs which impair metabolism of serotonin (including Monoamine Oxidase Inhibitors [MAOIs]). This may occur within the recommended dose. Serotonin syndrome may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular abnormalities (e.g. hyperreflexia, incoordination, rigidity), and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea).

If serotonin syndrome is suspected, rapid discontinuation of fentanyl should be considered.

Paediatric population:

Techniques that involve analgesia in a spontaneously breathing child should only be used as part of an anaesthetic technique, or given as part of a sedation/analgesia technique with experienced personnel in an environment that can manage sudden chest wall rigidity requiring intubation, or apnoea requiring airway support.

1. Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response. If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response. Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

2. Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue if serotonin syndrome is suspected.

3. Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

4. Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility. (See Postmarketing Experience)

Drug dependence and potential for abuse

Tolerance, physical dependence, and psychological dependence may develop upon repeated administration of opioids. Risks are increased in patients with a personal history of substance abuse (including drug or alcohol abuse or addiction).

Withdrawal syndrome

Repeated administration at short term intervals for prolonged periods may result in the development of withdrawal syndrome after cessation of therapy, which may manifest by the occurrence of the

following side effects: nausea, vomiting, diarrhoea, anxiety, chills, tremor, and sweating.

DRUG INTERACTIONS

Effect of other drugs on fentanyl

Drugs such as barbiturates, benzodiazepines, neuroleptics, halogenic gases, and other non-selective CNS depressants (e.g. alcohol) may potentiate the respiratory depression of narcotics. When patients have received such drugs, the dose of fentanyl required will be less than usual. Fentanyl, a high clearance drug, is rapidly and extensively metabolized mainly by CYP3A4. Itraconazole (a potent CYP3A4 inhibitor) at 200 mg/day given orally for 4 days had no significant effect on the pharmacokinetics of I.V. fentanyl.

Co-administration of fluconazole or voriconazole and fentanyl may result in an increased exposure to fentanyl.

Oral ritonavir (one of the most potent CYP3A4 inhibitors) reduced the clearance of I.V. fentanyl by two thirds; however, peak plasma concentrations after a single dose of I.V. fentanyl were not affected. When fentanyl is used in a single dose, the concomitant use of potent CYP3A4 inhibitors such as ritonavir requires special patient care and observation.

With continuous treatment of fentanyl and concomitant administration of CYP3A4 inhibitors a dose reduction of fentanyl may be required to avoid accumulation of fentanyl, which may increase the risk of prolonged or delayed respiratory depression.

It is usually recommended to discontinue MAO-inhibitors 2 weeks prior to any surgical or anesthetic procedure. However, several reports describe the uneventful use of fentanyl during surgical or anaesthetic procedures in patients on MAO- inhibitors.

When fentanyl is used in combination with non-vagolytic muscle relaxants, bradycardia and possibly asystole may occur.

Concomitant use of fentanyl and droperidol can result in higher incidence of hypotension. Pretreatment with, or concurrent administration of, cimetidine may increase plasma levels of fentanyl, when repeated doses of both drugs are used.

Bradycardia may be intensified by pretreatment with, or concurrent use of, drugs such as beta-blockers, suxamethonium, halothane, vecuronium, which may themselves cause bradycardia. Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Co-administration of fentanyl with a serotonergic agent, such as a Selective Serotonin Re-uptake Inhibitor (SSRI) or a Serotonin Norepinephrine Re-uptake Inhibitor (SNRI) or a Monoamine Oxidase Inhibitor (MAOI), may increase the risk of serotonin syndrome, a potentially life-threatening condition. Effect of fentanyl on other drugs

Following the administration of fentanyl, the dose of other CNS depressant drugs should be reduced. Plasma concentration of etomidate increased considerably (by a factor 2 to 3) when combined with fentanyl. The total plasma clearance and volume of distribution of etomidate are decreased by a factor 2 to 3 without a change in half-life when administered with fentanyl. Simultaneous administration of fentanyl and intravenous midazolam results in an increase in the terminal plasma half-life and a reduction in the plasma clearance of midazolam. When these drugs are co-administered with fentanyl their dose may need to be reduced.

Sedative medicines such as benzodiazepines or related drugs:

1. Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

USE IN SPECIFIC POPULATIONS

Pregnancy

There are no adequate data from the use of fentanyl in pregnant women. Fentanyl can cross the placenta in early pregnancy. Studies in animals have shown some reproductive toxicity. The potential risk for humans is unknown.

I.M. or I.V. administration during childbirth, (including caesarean section) is not recommended because Fentanyl crosses the placenta and because the foetal respiratory centre is particularly sensitive to opiates. If fentanyl is nevertheless administered, an antidote for the child should always be at hand. Breast-feeding

Fentanyl is excreted into human milk. Therefore, nursing is not recommended for 24 hours following the administration of this drug. The risk/benefit of breast-feeding following fentanyl administration should be considered.

ADVERSE REACTIONS

The following table displays ADRs that have been reported with the use of fentanyl:

System Organ Class	Adverse Drug Reactions			
	Frequency Category			
	Very Common	Common	Uncommon	Not Known
Immune System Disorders				Hypersensitivity (such as anaphylactic shock, anaphylactic reaction, urticaria)
Psychiatric Disorders			Euphoric mood	Delirium
Nervous System Disorders		Dyskinesia; Sedation; Dizziness	Headache	Convulsions; Loss of consciousness; Myoclonus; Hyperalgesia
Eye Disorders		Visual disturbance		
Cardiac Disorders		Bradycardia; Tachycardia; Arrhythmia		Cardiac arrest
Vascular Disorders		Hypotension; Hypertension; Vein pain	Phlebitis; Blood pressure fluctuation	
Respiratory, Thoracic and Mediastinal Disorders		Laryngospasm; Bronchospasm; Apnoea	Hyperventilation; Hiccups	Respiratory depression
Gastrointestinal Disorders	Nausea; Vomiting			
Skin and Subcutaneous Tissue Disorders		Dermatitis allergic		Pruritus

Musculoskeletal and Connective Tissue Disorder	Muscle Rigidity (which may also involve the thoracic muscles)			
General Disorders and Administration Site Conditions			Chills; Hypothermia	Drug withdrawal syndrome
Injury, Poisoning and Procedural Complications		Confusion postoperative; Anesthetic complication neurological	Airway complication of anaesthesia; Agitation postoperative; Procedural complication	

When a neuroleptic is used with fentanyl, the following adverse reactions may be observed: chills and/or shivering, restlessness, postoperative hallucinatory episodes and extrapyramidal symptoms.

OVERDOSAGE

Symptoms

An overdosage of fentanyl manifests itself as an extension of its pharmacologic actions. Depending on the individual sensitivity, the clinical picture is determined primarily by the degree of respiratory depression, which varies from bradypnoea to apnoea.

Management

In the presence of hypoventilation or apnoea, oxygen should be administered and respiration should be assisted or controlled as indicated. A specific narcotic antagonist, such as naloxone, should be used as indicated to control respiratory depression. This does not preclude the use of more immediate countermeasures. The respiratory depression may last longer than the effect of the antagonist; additional doses of the latter may therefore be required.

If depressed respiration is associated with muscular rigidity, an intravenous neuromuscular blocking agent might be required to facilitate assisted or controlled respiration.

The patient should be carefully observed; body warmth and adequate fluid intake should be maintained. If hypotension is severe or if it persists, the possibility of hypovolaemia should be considered and, if present, it should be controlled with appropriate parenteral fluid administration.

Effects on Ability to Drive and Use Machines

Fentanyl has a moderate influence on the ability to drive and use machines. Patients should only drive or operate a machine if sufficient time has elapsed after the administration of fentanyl.

This medicine can impair cognitive function and can affect a patient's ability to drive safely. When prescribing this medicine, patients should be told:

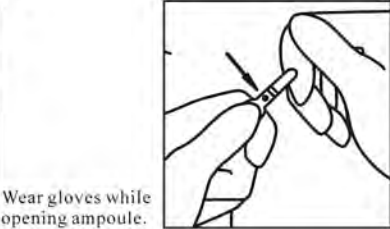
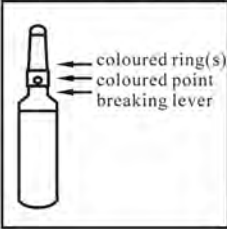
- The medicine is likely to affect your ability to drive
- Do not drive until you know how the medicine affects you
- It is an offence to drive while under the influence of this medicine
- However, you would not be committing an offence (called 'statutory defence') if:
 - The medicine has been prescribed to treat a medical or dental problem and
 - You have taken it according to the instructions given by the prescriber and in the information provided with the medicine and
 - It was not affecting your ability to drive safely

INCOMPATIBILITIES

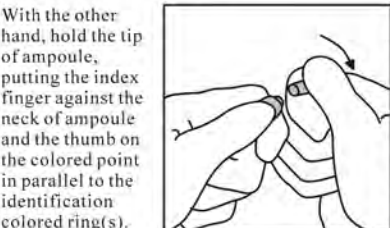
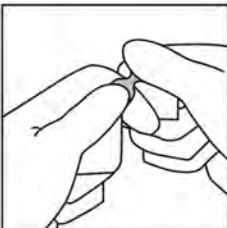
The injectable solution must not be mixed with other products.

If desired, Fentwell 0.1mg/2ml Solution for Injection may be mixed with sodium chloride or glucose intravenous infusions. Such dilutions are compatible with plastic infusion sets. These should be used within 24 hours of preparation.

Instructions for Use and Handling



Maintain the ampoule between thumb and index finger, leaving the tip of the ampoule free.



With the other hand, hold the tip of ampoule, putting the index finger against the neck of ampoule and the thumb on the colored point in parallel to the identification colored ring(s).

Accidental dermal exposure should be treated by rinsing the affected area with water. Avoid use of soap, alcohol, and other cleaning materials that may cause chemical or physical abrasions to the skin.

STORAGE CONDITION

Store at temperature below 30°C. Well-sealed. Protect from light.

After dilution:

Fentwell is stable for 24 hours at room temperature (below 30°C) after reconstitution and further dilution with sodium chloride or glucose injection.

PACKAGING AVAILABLE

10 x 2ml

The packaging form is 10 ampoules/packet. The ampoule is made of neutral borosilicate glass containing 2ml of Fentwell solution for each ampoule.

100 x 2ml

The packaging form is 10 ampoules/packet, then 10 packets/box. The ampoule is made of neutral borosilicate glass containing 2ml of Fentwell solution for each ampoule.

MANUFACTURER

Company Name: Yichang Humanwell Pharmaceutical Co., Ltd.

Manufacture Address: No. 19, Dalian Road, Yichang Developing Zone, Yichang, Hubei, China

DATE OF REVISION

October 2021



宜昌人福药业
YICHANG HUMANWELL